# Contents



# Global Flood Hydrotectonics

### A Quantitative Theory of Rapid Continental Reorganization

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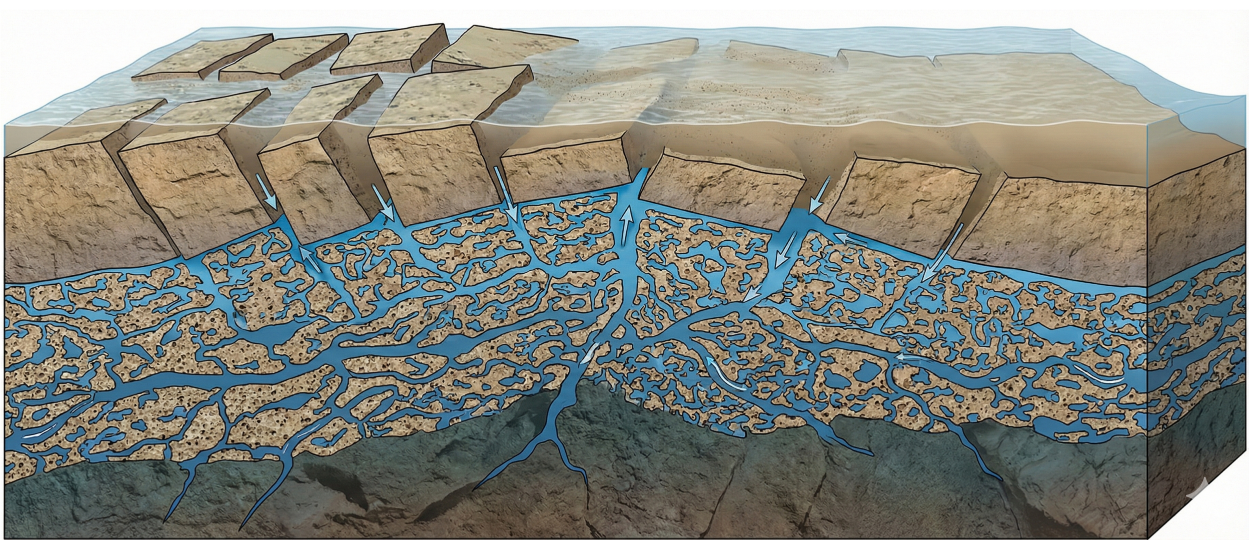

Figure 1: Channeled-Porosity Architecture

*Figure: Channeled-porosity architecture enabling sustained pore pressure during Stage 2 collapse. High-permeability channels supply water faster than low-permeability matrix can drain it.*

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**Date:** December 2025

**Status:** Preprint - not peer reviewed; released for open scientific evaluation and critique

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## How to Read This Document

This document should be read as a *working research programme*, not as a settled or comprehensive theory. Its purpose is to test physical plausibility under constrained assumptions, not to assert historical certainty or claim empirical confirmation. Some boundary conditions are stipulated rather than derived, and these are identified explicitly; subsequent analysis examines whether the proposed mechanisms operate coherently within known physical laws given those conditions. Where quantitative estimates are presented, they are order-of-magnitude unless otherwise stated. Predictions are distinguished from reinterpretations, and unresolved issues are acknowledged without claim of resolution. Readers are encouraged to evaluate the model on its internal consistency, energy and heat accounting, and exposure to potential falsification, rather than on agreement with its underlying metaphysical or interpretive commitments.

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## Executive Summary

This preprint presents *Global Flood Hydrotectonics* as a working physical research programme exploring whether rapid, large-scale continental reorganization during a global flood event can be made energetically and thermodynamically coherent under known physical laws. The model addresses the longstanding "heat problem" associated with catastrophic plate motion by proposing a hydraulic rather than mantle-viscous driving mechanism, in which continental blocks undergo hydroplaning along fluid-saturated detachment horizons under conditions of near-lithostatic pore pressure.

**Core Innovation:** Using Terzaghi's effective stress principle, continental blocks move by hydroplaning on thin water films along shallow detachment horizons, not by accelerated mantle convection. Energy dissipation occurs in water rather than through viscous mantle shear, reducing heat generation by two orders of magnitude.

**Key Results (Order-of-Magnitude):** - Heat flux: ~20 W/m$^2$ (vs. ~600 W/m$^2$ in conventional catastrophic models) - Global temperature rise: <1 K (vs. hundreds of Kelvin) - Energy source: Gravitational potential energy (~10$^{25}$ J) stored at Creation

**The Three-Stage Framework:**

| Stage | Name | Mechanism | Duration |
|---|---|---|---|
| 0 | Fiat/Creation | Stipulated initial conditions | Instantaneous |
| 1 | Stasis | Metastable equilibrium under physical law | Antediluvian era |
| 2 | Discharge | Hydraulic collapse triggered by seal failure | ~1 year |

The framework explicitly separates novel, discriminating predictions from post-hoc reinterpretations of existing observations. Quantitative estimates, reproducible calculations, and known uncertainties are presented transparently. The work does not claim confirmation, consensus, or completeness, but aims to establish whether the proposed mechanism constitutes a physically plausible research programme exposed to empirical risk and amenable to further testing, refinement, or falsification.

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## 1. The Heat Problem

Any model proposing rapid, global-scale tectonic reorganization must confront the energy budget. In standard plate tectonics, continental drift occurs at centimeters per year. Accelerating this to kilometers per day generates heat that scales with velocity squared.

**The Physical Constraint:** For viscous flow, dissipation increases as $v^2$. A hundred-fold acceleration produces ten-thousand-fold increase in heat generation. This heat exceeds Earth's radiative capacity, causing surface temperatures sufficient to vaporize oceans.

**The Conventional Objection:** Moving continental blocks thousands of kilometers in one year, using conventional friction assumptions, generates ~600 W/m$^2$ heat flux - nearly half the power of sunlight. This would be instantly lethal.

**The Hydrotectonic Solution:** The model achieves a 100× reduction in heat generation through Terzaghi's effective stress principle. When pore pressure approaches lithostatic stress, effective normal stress drops to ~1% of full lithostatic, and friction becomes negligible.

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## 2. The Three-Stage Framework

### Stage 0: Fiat Initial Conditions

The biblical text provides a boundary condition: "Let the dry land appear" (Genesis 1:9). This implies vertical separation - continents elevated while ocean basins sink.

**Physical Translation:** To raise buoyant continents to ~2 km average freeboard while maintaining isostatic equilibrium requires dense "ballast" positioned deep in the mantle. The model interprets cold slabs observed in seismic tomography as consistent with a **Created Ballast** hypothesis, rather than requiring long-term subduction.

**The Gravitational Battery:** Stage 0 emplacement stores ~10$^{25}$ J of gravitational potential energy - sufficient to power Stage 2 reorganization without requiring thermal convection.

**Methodological Note:** Stage 0 invokes supernatural causation for initial conditions only. This is methodologically parallel to cosmological models that accept Big Bang initial conditions while modeling subsequent evolution through natural law.

### Stage 1: Antediluvian Stasis

Following Stage 0, Earth existed in a high-energy but stable configuration. Stability derived from hydraulic seals - low-permeability layers that isolated high-pressure fluid reservoirs.

**The Pre-Flood Crustal Architecture:** - Water-saturated lithosphere with distributed aquifers - Interconnected fracture networks and fault-bounded aquifers - Overpressured compartments approaching lithostatic pressure - Serpentinized upper mantle holding 10-15 wt% water

**The Channeled-Porosity Architecture:** A critical feature - not sealed compartments but an open-flow system with continuous water supply. This determines which physics governs Stage 2: submarine hydroplaning physics rather than fault-valve drainage.

### Mineralogical Evidence for Prior Water Mobility

The model requires that substantially more water was mobile in the past than is accessible today. This is not merely stipulated - it is recorded in Earth's mineral inventory.

**The Ringwoodite Evidence:** In 2014, Pearson et al. reported a hydrous ringwoodite inclusion trapped within a diamond from the mantle transition zone (~520-660 km depth). This single crystal contained ~1.5 wt% $H_2O$ - direct evidence that the transition zone has hosted significant water. Ringwoodite can only incorporate water during formation under elevated water fugacity. The water now locked in this mineral was once mobile.

| Mineral / Phase | Depth Range | $H_2O$ Capacity | Hydration Requirement | Relevance |
|---|---|---|---|---|
| **Ringwoodite** | ~520-660 km | ~1-3 wt% | Requires prior mobile water at high pressure | Demonstrates large-scale sequestration following high-pressure hydration |
| **Wadsleyite** | ~410-520 km | Up to ~3 wt% | Hydrates only under elevated water fugacity | Indicates extensive water-rich regime preceding present conditions |
| **Serpentine group** | Upper mantle/lithosphere | ~10-15 wt% | Forms by hydration of peridotite via liquid water | Records large-scale infiltration consistent with transient fluid mobilization |
| **Brucite** | Upper mantle | ~30 wt% (local) | Requires excess water during hydration | Temporary water sink during rapid hydration events |

| Mineral / Phase | Depth Range | $H_2O$ Capacity | Hydration Requirement | Relevance |
|---|---|---|---|---|
| **Chlorite** | Lower crust to upper mantle | ~10-13 wt% | Forms under water-rich metamorphic conditions | Supports widespread fluid involvement beyond isolated faults |
| **NAMs (olivine, pyroxene, garnet)** | Crust to deep mantle | Tens-hundreds ppm | Requires elevated water fugacity | Indicates pervasive hydration exceeding present-day background |
| **DHMS phases** | Transition zone to lower mantle | Up to several wt% | Stable only at high pressure with abundant water | Extends plausible depth of water sequestration |

*The widespread occurrence of hydrous and hydrogen-bearing mantle minerals is interpreted here not merely as storage capacity, but as physical evidence of a prior regime characterized by substantially greater water mobility and elevated pore pressures, followed by large-scale sequestration during system relaxation.*

### Hydration as Dissipative Endpoint

The hydrotectonic collapse provides the conditions necessary for rapid, large-scale mineral hydration. During Stage 2, pore pressures approach lithostatic, effective stress collapses, and water is forced into grain boundaries, fracture networks, and reaction fronts under near-failure conditions. This is not slow percolation but forced infiltration at system scale.

Elevated temperature (from deformation and friction) combined with sustained high pressure dramatically accelerates hydration kinetics. Serpentinization, brucite and chlorite formation, hydrogen incorporation into nominally anhydrous minerals, and hydration of transition-zone phases all proceed rapidly when water activity is high and stress barriers are low - precisely the conditions the model provides.

As the event progresses and hydraulic energy dissipates, water transitions from free mobile fluid to chemically bound $OH^-$ and $H^+$ in mineral lattices. Mineral hydration thus functions as a dissipative endpoint: a water sink that absorbs mobile fluid during pressure decay. Once pore pressures fall, fracture permeability collapses, and temperature gradients normalize, the system becomes mechanically stiff and hydraulically isolated. Bound water is no longer mobile. The modern low-flux Earth regime begins.

**Testable signatures:** This mechanism predicts texturally rapid hydration fronts, minimal overprinting, and spatial mismatch between hydration extent and long-term subduction geometry. These are measurable. If absent, the model fails. If present, slow-only explanations weaken.

**Stage 2: The Flood Event (Discharge)**

**Trigger:** Large meteor impacts generated shockwaves that fractured aquitard layers, initiating cascading seal failure across global scales.

**Mechanism - Effective Stress Collapse:**

As pore pressure ($P_p$) approaches lithostatic stress ($\sigma_L$):

$$\sigma_{eff} = \sigma_L - P_p \to 0$$

When effective stress drops below ~1% of lithostatic, friction becomes negligible. Crustal blocks accelerate under gravity.

**Hydroplaning:** Continental blocks moved on thin water films along shallow detachment horizons (~50 km depth). Velocities of tens to hundreds of meters per hour are consistent with the energy budget.

**Observational Analog:** Submarine landslide hydroplaning - mass transport deposits travel extraordinary distances on slopes as gentle as 1°, far exceeding conventional friction predictions. The mechanism has been documented in laboratory experiments and field observations.

**Termination:** Stage 2 ended as the hydraulic system exhausted itself. Pore pressures dropped, effective stress recovered, and plate velocities decayed to the modern centimeters per year.

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## 3. Key Mechanisms

### 3.1 Heat Dissipation: Hypercanes

Frictional warming raised ocean temperatures into the Hypercane-forming regime (>40°C). Hypercanes: - Pump heat from ocean surface into the stratosphere - Radiate thermal energy into space - Accelerate heat dissipation beyond passive radiation

### 3.2 Sediment Transport: Turbidity Currents

Stage 2 sedimentation operated through two mechanisms:

**Shallow Environments:** Hypercane-driven waves eroded exposed rock, producing coarse clastics and basal unconformities (e.g., Great Unconformity).

**Deep Environments:** Sediment-laden water became denser than ambient seawater, driving turbidity currents - gravity-driven flows that cascaded into basins, producing graded bedding and Bouma sequences.

### 3.3 Fossil Distribution

Fossil order emerges from three non-temporal factors: 1. **Ecological zonation:** Pre-Flood organisms occupied elevation-based habitats 2. **Hydraulic sorting:** Organisms sort by size, density, and hydrodynamic properties 3. **Depositional geometry:** Basin-scale processes concentrate different organisms in different facies

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## 4. Energy Budget Summary

### Available Energy

| Component | Energy (J) |
|---|---|
| Single continental block settling 1 km | $10^{24}$ |
| 10 major blocks settling | $10^{25}$ |

### Energy Partitioning (from numerical simulation)

| Sink | Percentage | Energy (J) |
|---|---|---|
| Frictional dissipation | 5.0% | $3.2 \times 10^{23}$ |
| Viscous dissipation | 0.5% | $3.2 \times 10^{22}$ |
| Seismic radiation | 0.25% | $1.6 \times 10^{22}$ |
| Residual PE (blocks in new equilibrium) | 94.2% | $6.0 \times 10^{24}$ |

### Heat Flux Comparison

| Model | Heat Flux | Global Warming |
|---|---|---|
| Conventional catastrophic | ~600 W/m$^2$ | Lethal (hundreds K) |
| Hydrotectonic | ~20 W/m$^2$ | <1 K |

The 100× reduction arises entirely from pore pressure effects (Terzaghi's principle).

### Hydrotectonic vs. Geothermal Work

Although magmatic and geothermal processes likely contributed work during Stage 2, the dominant mechanical driver in this model is hydrotectonic collapse. The comparison:

| Energy Source | Flood-Year Energy (J) | Basis |
|---|---|---|
| Gravitational PE (hydrotectonic) | ~$10^{25}$ | 10 major blocks settling ~1 km |
| Geothermal flux (integrated) | ~$1.4 \times 10^{21}$ | 0.09 W/m$^2$ × Earth surface × 1 year |

The ratio is approximately **10,000:1**. Gravitational potential energy released by rapid block reorganization couples primarily into high-pressure fluid flow and low-effective-stress detachment motion.

Geothermal flux is a secondary contributor to the flood-year energy budget relative to hydraulically mediated mechanical work.

**Falsifiability:** This dominance claim is testable. If independent constraints show that required hydrotectonic dissipation is comparable to or less than plausible flood-year geothermal/magmatic work, the model's central driver assumption fails.

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## 5. Testable Predictions

### Novel Predictions (Genuinely Risky)

**Structural Unconformity:** High-resolution seismic tomography should reveal a texture discontinuity between chaotic, reworked upper lithosphere (0-200 km) and coherent, undisturbed deep mantle (>400 km).

**Fluid-Dominated Detachments:** Large-scale detachment horizons should show predominantly fluid-related fabrics with minimal pseudotachylite (frictional melt).

**Basin Margin Megabreccia:** Chaotic megabreccia at basin boundaries with multi-lithology mixing, fluid-flow textures, and rapid emplacement signatures.

### Post-Hoc Accommodations (Not Confirmations)

- Transition zone water content (Pearson et al., 2014) - known before model developed
- Cold slabs at depth (van der Hilst, Fukao) - model reinterprets, does not predict

### Discriminating Predictions

- Isotopic signatures of deep water (surface-derived vs. primordial)
- Thermal gradient analysis distinguishing "emplaced cold" from "slowly subducted cold"

### Speculative Implication: Geomagnetic Excursions

The extreme hydrotectonic activity proposed for Stage 2 entails rapid global mass redistribution, transient thermal boundary changes at the core-mantle interface, and sustained large-scale conductive fluid motion. Each of these processes is independently known to influence geomagnetic field behavior. Acting concurrently, they provide a plausible mechanism for repeated geomagnetic excursions or rapid directional instabilities during the flood interval, without requiring long-term dynamo shutdown or stable polarity reversals. This interpretation predicts compressed paleomagnetic sequences characterized by abrupt directional variability and intensity collapse rather than prolonged transitional states.

*Note: This is among the model's more speculative components. See I-0004 for detailed exploration.*

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## 6. Limitations and Open Questions

This model acknowledges unresolved issues:

1. **Energy partitioning uncertainties:** Seismic efficiency and viscous dissipation fractions are order-of-magnitude estimates
2. **Scale extrapolation:** Submarine hydroplaning observed at 100s km; model requires 1000s km
3. **Pore pressure maintenance:** Mechanism plausible via channeled-porosity, but not directly demonstrated at crustal scale
4. **Temporal dynamics:** Steady-state assumed; start-up transients not modeled
5. **Spatial heterogeneity:** Model treats blocks uniformly; actual motion would be variable

These are modeling challenges, not physical impossibilities.

**Open Exploration Directions**

The following questions define the empirical risk and future testability of this research programme:

1. **Timescale compression:** Can hydrotectonic, hydration, and depositional processes proceed at flood-year timescales without violating chemical, mechanical, or magnetic relaxation limits?

2. **Permeability evolution:** Can global-scale permeability increase during failure and collapse fast enough to trap fluids post-event?

3. **Energy partitioning robustness:** Is hydrotectonic dominance over geothermal contributions stable across reasonable parameter bounds?

4. **Water budget quantification:** Is inferred once-mobile water volume consistent with mineral hydration capacities and mobilization efficiencies?

5. **Mineralogical discrimination:** Do hydration textures or isotopic signatures distinguish rapid global hydration from prolonged subduction-driven hydration?

6. **Sediment transport:** Can hydrotectonic-driven turbidity reproduce observed volumes and sorting patterns without long timescales?

7. **Geomagnetic excursions:** Can rapid mass redistribution and fluid flow generate multiple excursions within a compressed interval?

8. **Falsifiability targets:** Which single observation would most decisively falsify the framework?

**Cross-cutting constraint:** Timescale compression (#1) is the highest-risk issue because it affects hydration kinetics, sediment transport, and paleomagnetic remanence acquisition simultaneously. If any one of these processes demonstrably requires timescales incompatible with the model, the framework fails at multiple points.

*See I-0005 for detailed sub-issues and phase mapping.*

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## 7. Theoretical Framework

The model operates within the **Christian Designism** research programme:

1. **Literal Programmatic Intervention (LPI):** God acts through designed, law-governed processes with scheduled interventions at specific historical points

2. **Fiat Initial Conditions:** Creation events establish boundary conditions that subsequent natural processes operate within
3. **Physical Plausibility:** Between interventions, the system operates according to discoverable physical laws

**What we model:** Stages 1 and 2 operate through natural physical processes - subject to quantitative analysis, prediction, and potential falsification.

**What we stipulate:** Stage 0 initial conditions are specified by biblical testimony, not derived from physical law - analogous to cosmological models accepting Big Bang initial conditions.

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## 8. Reference Materials

### Primary Documents

| Document | Location | Description |
|---|---|---|
| Full Model (v2.5) | `theory/papers/20251217-hydrotectonic-model-collapse.md` | Complete theory with all appendices |
| Energy Simulation | `notebooks/20251217_energy_partitioning_simulation.ipynb` | Reproducible numerical calculations |
| Gap Analysis | `notebooks/20251217_gap_analysis.ipynb` | Pressure stability and scale-up analysis |

### Appendices (in Full Model)

- **A:** Gravitational Potential Energy Budget
- **B:** Heat Dissipation and Thermal Budget (shear-stress derivation, energy partitioning)
- **C:** Hypercane Physics and Heat Transport
- **D:** Driving Forces and Block Velocity
- **E:** Earth-System Context for Hydraulic Collapse
- **F:** Darcy Flow Calculations for Channeled-Porosity Architecture
- **G:** Lubrication Theory Analysis (Reynolds equation, film stability)
- **H:** Gap Analysis (diffusion timescales, pressure stability, scale-up)

### Supporting Materials

| Folder | Contents |
|---|---|
| `theory/calculations/` | Detailed calculations (driving forces, kinetic energy, steam) |
| `theory/literature/` | Literature reviews and citation validation |
| `theory/defense/` | Responses to objections and reviewer feedback |
| `theory/figures/` | Generated figures from simulations |

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## 9. Key References


Anderson, D.L. (2007). New Theory of the Earth. Cambridge University Press.

De Blasio, F.V., et al. (2004). Hydroplaning and submarine debris flows. Journal of Geophysical Research, 109.

Emanuel, K.A. (1994). Hypercanes: A possible link in global extinction scenarios. Journal of Geophysical Research.

Flemings, P.B., et al. (2008). Pore pressure penetrometers document high overpressure near the seafloor. Earth and Planetary Science Letters.

Fukao, Y., and Obayashi, M. (2013). Subducted slabs stagnant above, penetrating through, and trapped below the 660 km discontinuity. Journal of Geophysical Research.

Mohrig, D., et al. (1998). Hydroplaning of subaqueous debris flows. GSA Bulletin.

Moore, J.C., and Saffer, D. (2001). Updip limit of the seismogenic zone. Geology.

Pearson, D.G., et al. (2014). Hydrous mantle transition zone indicated by ringwoodite included within diamond. Nature.

van der Hilst, R.D., et al. (1997). Evidence for deep mantle circulation from global tomography. Nature.

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## Appendix: Research Programme and Open Problems

This appendix summarizes the structured research agenda embedded in this paper. Issues are ranked by expected reviewer attack likelihood.

### High-Priority Constraints (Attack Likely)

| Rank | Issue | Why Vulnerable |
|---|---|---|
| 1 | **Timescale compression** | Cross-cutting: affects hydration, sedimentation, paleomagnetism. Most common objection. |
| 2 | **Permeability evolution** | Core mechanism depends on it. No validated constitutive law exists. |
| 3 | **Energy partitioning** | Underpins heat problem solution. Sensitivity to assumptions will be probed. |

### Medium-Priority Constraints (Engagement Possible)

| Rank | Issue | Why Relevant |
|---|---|---|
| 4 | **Water budget** | "Where did the water go?" is intuitive. Mineral sequestration argument is less familiar. |
| 5 | **Mineralogical discrimination** | Empirical: could yield discriminating evidence if pursued. |
| 6 | **Sediment transport** | Connects deep mechanism to surface geology. Reviewers may demand observational links. |

**Lower-Priority Constraints (Derivative)**

| Rank | Issue | Notes |
|---|---|---|
| 7 | **Geomagnetic excursions** | Marked speculative. Defended by I-0004. Not core. |
| 8 | **Falsifiability targets** | Actually strengthens paper. Unlikely to be attacked. |

### Core Claims (Not Negotiable)

These claims define the research programme's hard core:

1. Continental motion via hydraulic decoupling, not accelerated mantle convection
2. Gravitational PE as primary energy source (~$10^{25}$ J)
3. Terzaghi effective stress principle driving friction collapse
4. Stage 0 stipulated initial conditions (boundary condition, not modeled)

### Protective Belt (Modifiable)

These auxiliary hypotheses can be adjusted without abandoning the core:

- Specific trigger mechanism (meteors vs. other)
- Exact detachment depth
- Hypercane efficiency parameters
- Episodic vs. continuous motion
- Permeability evolution law (once constrained)

### Discriminating Predictions

The programme advances if these are confirmed; it degenerates if they fail:

- Structural unconformity at ~200 km depth
- Fluid-dominated detachment fabrics (not pseudotachylite)
- Rapid-hydration mineral textures
- Compressed paleomagnetic sequences (if geomagnetic claim pursued)

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## Document Information

**Version History:** - v1.0 (Nov 2024): Original hydrotectonic collapse model - v2.0 (Dec 2024): Three-stage framework addressing deep mantle paradox - v2.5 (Dec 2025): Gap analysis edition with numerical simulation

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**Repository:** github.com/jdlongmire/global-flood-hydrotectonic-model



**Citation:** > Longmire, J.D. (2025). Global Flood Hydrotectonics: A Quantitative Theory of Rapid Continental Reorganization (Version 2.5). Zenodo.

**Collaboration:** This is an open research programme built on reproducible Jupyter notebooks. Contributions welcome - especially computational modeling of pressure stability, permeability evolution, thermal feedback, and discriminating predictions. Open an issue or contact the author.